DealMatch — Updated Entities and Foreign Keys

Latest model: Category/Subcategory-based personalization. Interest-related entities are removed.

Entity	Attributes	Foreign Keys
User	UserId PK; Name; Email UNIQUE; PasswordHash; Phone; AccountStatus; CreatedAt; UpdatedAt	
Role	RoleId PK; Name; Description	
Permission	PermissionId PK; Name; Description	
UserRole	UserId PK/FK; RoleId PK/FK	UserId → User.UserId; RoleId → Role.RoleId
RolePermission	RoleId PK/FK; PermissionId PK/FK	RoleId → Role.RoleId; PermissionId → Permission.PermissionId
Merchant	MerchantId PK; BusinessName; BusinessEmail; BusinessPhone; BusinessAddress; Description; AccountStatus; VerificationStatus	
MerchantUser	MerchantId PK/FK; UserId PK/FK	MerchantId → Merchant.MerchantId; UserId → User.UserId
Category	CategoryId PK; Cat_Name; Description; ParentCategoryId FK nullable; IsActive; DisplayOrder	ParentCategoryId → Category.CategoryId
UserCategoryPreference	UserId PK/FK; CategoryId PK/FK; SelectedAt	UserId → User.UserId; CategoryId → Category.CategoryId
Deal	DealId PK; MerchantId FK; CategoryId FK; Title; Description; OriginalPrice; DealPrice; Currency; StartAt; EndAt; MaxClaimsPerUser; MaxClaimsTotal; TermsAndConditions; ApprovalStatus; PublicationStatus	MerchantId → Merchant.MerchantId; CategoryId → Category.CategoryId
DealMedia	MediaId PK; DealId FK; MediaUrl; MediaType; DisplayOrder	DealId → Deal.DealId
Inventory	DealId PK/FK; AvailableQuantity; ReservedQuantity; ...	DealId → Deal.DealId
InventoryMovement	InventoryMovementId PK; DealId FK; QuantityChange; Reason; ActorUserId FK nullable; CreatedAt	DealId → Deal.DealId; ActorUserId → User.UserId
Favorite	UserId PK/FK; DealId PK/FK; AddedAt	UserId → User.UserId; DealId → Deal.DealId
Notification	NotificationId PK; RecipientUserId FK; DealId FK nullable; OrderId FK nullable; Type; Title; Message; IsRead; CreatedAt; ReadAt nullable	RecipientUserId → User.UserId; DealId → Deal.DealId; OrderId → Order.OrderId
Order	OrderId PK; UserId FK; OrderNumber UNIQUE; OrderStatus; Currency; SubtotalAmount; DiscountAmount; TaxAmount nullable; TotalAmount; BillingName; BillingEmail; BillingAddress nullable; CreatedAt; UpdatedAt; CancelledAt nullable	UserId → User.UserId
OrderItem	OrderItemId PK; OrderId FK; DealId FK; MerchantId FK; DealTitleSnapshot; Quantity; UnitOriginalPrice; UnitDealPrice; LineSubtotal; DiscountAmount; LineTotal; ItemStatus; CreatedAt; UpdatedAt	OrderId → Order.OrderId; DealId → Deal.DealId; MerchantId → Merchant.MerchantId
Payment	PaymentId PK; OrderId FK; ProviderName; ProviderPaymentReference; Amount; Currency; PaymentStatus; PaymentMethodType; CreatedAt; AuthorizedAt; PaidAt; FailedAt; FailureReason	OrderId → Order.OrderId
PaymentTransaction	PaymentTransactionId PK; PaymentId FK; ProviderTransactionReference; TransactionType; Amount; Currency; TransactionStatus; CreatedAt; CompletedAt; ProviderResponseCode	PaymentId → Payment.PaymentId
Refund	RefundId PK; OrderId FK; PaymentId FK; RequestedByUserId FK; RefundAmount; Currency; Reason; RefundStatus; ProviderRefundReference; RequestedAt; ProcessedAt; ProcessedByUserId FK nullable	OrderId → Order.OrderId; PaymentId → Payment.PaymentId; RequestedByUserId → User.UserId; ProcessedByUserId → User.UserId
RefundItem	RefundItemId PK; RefundId FK; OrderItemId FK; Quantity; RefundAmount	RefundId → Refund.RefundId; OrderItemId → OrderItem.OrderItemId
Redemption	RedemptionId PK; OrderItemId FK; UserId FK; DealId FK; CouponCode UNIQUE; Status; IssuedAt; ExpiresAt; RedeemedAt; RedeemedByMerchantId FK nullable; CancelledAt; RefundedAt	OrderItemId → OrderItem.OrderItemId; UserId → User.UserId; DealId → Deal.DealId; RedeemedByMerchantId → Merchant.MerchantId
Commission	CommissionId PK; MerchantId FK; OrderItemId FK; RefundId FK nullable; SaleAmount; CommissionRate; CommissionAmount; Currency; Status; CreatedAt; DueAt; AdjustedAt; AdjustmentReason	MerchantId → Merchant.MerchantId; OrderItemId → OrderItem.OrderItemId; RefundId → Refund.RefundId
MerchantPayout	MerchantPayoutId PK; MerchantId FK; ProviderPayoutReference; Amount; Currency; PayoutStatus; PeriodStart; PeriodEnd; CreatedAt; PaidAt; FailureReason	MerchantId → Merchant.MerchantId
MerchantPayoutItem	MerchantPayoutItemId PK; MerchantPayoutId FK; CommissionId FK; Amount	MerchantPayoutId → MerchantPayout.MerchantPayoutId; CommissionId → Commission.CommissionId

Entity	Attributes	Foreign Keys
Review	ReviewId PK; UserId FK; DealId FK; OrderItemId FK nullable; Rating; Comment; ReviewStatus; CreatedAt; UpdatedAt; ModeratedByAdminId FK nullable; ModeratedAt nullable	UserId → User.UserId; DealId → Deal.DealId; OrderItemId → OrderItem.OrderItemId; ModeratedByAdminId → User.UserId
Complaint	ComplaintId PK; UserId FK; DealId FK nullable; MerchantId FK nullable; OrderId FK nullable; Description; ComplaintStatus; CreatedAt; ResolvedAt; AssignedAdminId FK nullable	UserId → User.UserId; DealId → Deal.DealId; MerchantId → Merchant.MerchantId; OrderId → Order.OrderId; AssignedAdminId → User.UserId
AuditLog	AuditLogId PK; ActorUserId FK; Action; EntityType; EntityId; CreatedAt; Details	ActorUserId → User.UserId

What Changed

• Removed Interest, UserInterest, DealInterest, and InterestCategory.

• Added UserCategoryPreference(UserId, CategoryId, SelectedAt).

• Category now supports hierarchy through Category.ParentCategoryId → Category.CategoryId.

• Personalization uses UserCategoryPreference matched to Deal.CategoryId.

• Notification keeps optional DealId and OrderId; no CategoryId is required.

• Order has no ShippingAmount; it uses SubtotalAmount, DiscountAmount, optional TaxAmount, and TotalAmount.

• Merchant–User remains many-to-many through MerchantUser.

Personalization path: User → UserCategoryPreference → Category/Subcategory ← Deal